KCE PATHWAYS · GRADE 3 · UNIT 4: PARTS AND PIECES · WEEK 17 · MATH

The Long Lane at Clover Creek Dairy — Teacher Lesson Plan & Answer Key

Learning objectives: Students place and name fractions on a number line from 0 to 1 by counting equal spaces (not tick marks), keep counting jumps past 1 on a number line from 0 to 2, recognize whole numbers written as fractions (like 4/4 = 1 and 8/4 = 2), and find equivalent fractions that land on the same point or cover the same amount of a same-size whole.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Clover Creek Dairy near Madison, Wisconsin, with Mrs. Jensen, Wen, Zuri and Tomas and the 1-mile farm lane.
2. Warm-Up, Part 1: Fractions of a Whole (8 pts, Week 16 skill): naming the shaded fraction of a bar and a fraction of a group of calves. If a student misses both, review "denominator = all the equal parts" before moving on.
3. Warm-Up, Part 2 (4 pts): a find-the-mistake item about counting the equal parts of a bar.
4. Picture It: The Long Lane at Clover Creek Dairy. The lane from the barn (0) to the pasture gate (1) is exactly 1 mile long. Ask: if we put posts at equal spaces, how could we name each post?
5. Word Wrangler: 6 vocab cards (number line, endpoint, equal parts, numerator, denominator, equivalent fractions). Have the student say each meaning back.
6. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W17.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
7. Fractions on a Number Line, Part 1 (8 pts): naming points A and B on a 0-to-1 line. Tip: have the student put a finger on 0 and count the jumps (spaces), not the tick marks.
8. Fractions on a Number Line, Part 2 (4 pts): a reasoning item about counting equal spaces.
9. Counting Past 1, Part 1: Jumps to the Creek (8 pts): points on a 0-to-2 line and whole numbers as fractions (how many fourths make 2 wholes). Tip: keep counting the jumps after 1, so 3/3 is 1 and 6/3 is 2.
10. Counting Past 1, Part 2 (4 pts): a reasoning item about fractions past 1 on the number line.
11. Same Spot, Different Names, Part 1 (8 pts): two number lines stacked (halves over fourths, thirds over sixths) and missing numerators. Tip: slide a ruler or the edge of a paper straight down to see which points line up.
12. Same Spot, Different Names, Part 2 (4 pts): does the pair of fractions land on the very same point?
13. Cheese Blocks, Part 1: Equal Amounts (8 pts): equivalent fractions with same-size blocks and strips (3/4 of a block in eighths; 4/6 in thirds). Tip: the wholes must be the same size before you compare.
14. Cheese Blocks, Part 2 (4 pts): a reasoning item about whether the parts cover the same amount.
15. Bonus Practice, Part 1: The Feed Bin (XP only, not scored).
16. Bonus Practice, Part 2: Points on the Lane (XP only, not scored).

17. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches equivalent fractions, Set 2 matches number-line vocabulary to meanings, Set 3 matches fractions to equal amounts.
18. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters, names, end mark, list commas); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
19. Speed Arena, Part 1 (XP only, not scored).
20. Speed Arena, Part 2 and Keyboard Corral (XP only, not scored).
21. Boss Battle, Part 1: The Last Post (10 pts, 5 each): a two-step fence-post problem and a find-the-mistake item about equivalent distances.
22. Boss Battle, Part 2: Closing the Gate (10 pts, 5 each): naming a point on the line and a which-explanation-is-correct item about wholes such as 2/2 and 4/4.
23. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W17.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
24. Results (Fence Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Farm Lane Hero!” with the congratulations message “Farm lane hero! You marked every post on the lane!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every farmhand gets stronger with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: FRACTIONS OF A WHOLE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: This bar is cut into equal parts. What fraction of the bar is shaded? Type it like 1/2.	**1/6** (also accepted: 1 / 6, one sixth, one-sixth, 1 sixth, a sixth)	The bar has 6 equal parts and 1 is shaded, so the shaded part is 1/6. The bottom number, 6, counts the equal parts.
w2_s0	Set 1: A farmer has 8 calves. The dark circles show the 3 calves that are black. What fraction of the calves are black?	**3/8** (also accepted: 3 / 8, three eighths, three-eighths, 3 eighths)	There are 8 calves in the set, and 3 are black. 3 out of 8 is 3/8.
w3_s0	Set 1: Find the mistake. Tomas says the shaded part of this bar is 1/4 because the bar has 4 parts. What went wrong? [The 4 parts are not equal, so the shaded part is not one fourth. / Nothing is wrong. The shaded part is one fourth. / He should have said one third. / He should count the white parts, so it is three fourths.]	**The 4 parts are not equal, so the shaded part is not one fourth.**	Fourths must be 4 equal parts. These parts are different sizes, so the shaded part is not 1/4.
w1_s1	Set 2: What fraction of this bar is shaded? Type it like 1/2.	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths)	The bar has 8 equal parts and 5 are shaded: 5/8.
w2_s1	Set 2: Two steps! A loaf of bread is cut into 6 equal slices. 4 slices are eaten. What fraction of the loaf is left?	**2/6** (also accepted: 2 / 6, two sixths, two-sixths, 2 sixths, 1/3, 1 / 3, one third, one-third, 1 third, a third)	Step 1: 6 − 4 = 2 slices are left. Step 2: 2 of the 6 equal slices is 2/6 of the loaf.

ID	Question	Answer	Why (explain to student)
w3_s1	Set 2: Which explanation is correct? What does the 3 in the fraction 2/3 tell you? [The whole is cut into 3 equal parts. / 3 parts are shaded. / There are 3 wholes. / It is the biggest part.]	**The whole is cut into 3 equal parts.**	The denominator, the bottom number, tells how many equal parts make one whole. In 2/3, the whole is cut into 3 equal parts.
w1_s2	Set 3: What fraction of this bar is shaded? Type it like 1/2.	**2/3** (also accepted: 2 / 3, two thirds, two-thirds, 2 thirds)	The bar has 3 equal parts and 2 are shaded: 2/3.
w2_s2	Set 3: Two steps! A carton holds 6 eggs. 4 of the eggs are brown. What fraction of the eggs are NOT brown?	**2/6** (also accepted: 2 / 6, two sixths, two-sixths, 2 sixths, 1/3, 1 / 3, one third, one-third, 1 third, a third)	Step 1: 6 − 4 = 2 eggs are not brown. Step 2: 2 out of 6 is 2/6.
w3_s2	Set 3: Find the mistake. A bar has 5 equal parts, and 3 are shaded. Zuri wrote 5/3. What went wrong? [She swapped the numbers. It should be three fifths, with 3 on top and 5 on the bottom. / Nothing is wrong. Five thirds is correct. / It should be two fifths. / It should be three eighths.]	**She swapped the numbers. It should be three fifths, with 3 on top and 5 on the bottom.**	The numerator, on top, counts the shaded parts (3). The denominator, on the bottom, counts all the equal parts (5). The fraction is 3/5.

FRACTIONS ON A NUMBER LINE (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1a_s0	Set 1: What fraction names point A? Type it like 1/2.	**1/6** (also accepted: 1 / 6, one sixth, one-sixth, 1 sixth, a sixth, 1/6 mile, 1/6 of a mile, 1/6 miles, 1 / 6 mile, 1 / 6 of a mile, 1 / 6 miles)	The line from 0 to 1 has 6 equal spaces, so each jump is 1/6. Point A is 1 jump from 0: 1/6.
p1b_s0	Set 1: What fraction names point B?	**5/8** (also accepted: 5 / 8, five eighths, five-eighths, 5 eighths, 5/8 mile, 5/8 of a mile, 5/8 miles, 5 / 8 mile, 5 / 8 of a mile, 5 / 8 miles)	There are 8 equal spaces from 0 to 1. Point B is 5 jumps from 0, so it is at 5/8.
p1c_s0	Set 1: Find the mistake. Tomas looks at a number line from 0 to 1 with 5 tick marks at equal spaces. He says each part is 1/5. What went wrong? [He counted tick marks. There are only 4 equal spaces, so each part is one fourth. / Nothing is wrong. Each part is one fifth. / Each part is one sixth. / Each part is 5 miles long.]	**He counted tick marks. There are only 4 equal spaces, so each part is one fourth.**	Count the spaces, not the tick marks. 5 tick marks make only 4 equal spaces, so each part is 1/4.
p1a_s1	Set 2: What fraction names point C?	**2/3** (also accepted: 2 / 3, two thirds, two-thirds, 2 thirds, 2/3 mile, 2/3 of a mile, 2/3 miles, 2 / 3 mile, 2 / 3 of a mile, 2 / 3 miles)	The line is cut into 3 equal spaces. Point C is 2 jumps from 0: 2/3.

ID	Question	Answer	Why (explain to student)
p1b_s1	Set 2: What fraction names point D?	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths, 4/6 mile, 4/6 of a mile, 4/6 miles, 4 / 6 mile, 4 / 6 of a mile, 4 / 6 miles, 2/3, 2 / 3, two thirds, two-thirds, 2 thirds, 2/3 mile, 2/3 of a mile, 2/3 miles, 2 / 3 mile, 2 / 3 of a mile, 2 / 3 miles)	There are 6 equal spaces, and point D is 4 jumps from 0, so it is at 4/6. (That is the same spot as 2/3.)
p1c_s1	Set 2: Which explanation is correct? How do you find 1/3 on a number line from 0 to 1? [Cut the space from 0 to 1 into 3 equal parts. The first mark after 0 is one third. / Put a mark 3 spaces past 1. / Draw 3 tick marks anywhere between 0 and 1. / Cut the line into 4 parts and use the third mark.]	**Cut the space from 0 to 1 into 3 equal parts. The first mark after 0 is one third.**	To show thirds, cut the space from 0 to 1 into 3 equal parts. The first mark after 0 is 1/3.
p1a_s2	Set 3: What fraction names point E?	**3/8** (also accepted: 3 / 8, three eighths, three-eighths, 3 eighths, 3/8 mile, 3/8 of a mile, 3/8 miles, 3 / 8 mile, 3 / 8 of a mile, 3 / 8 miles)	There are 8 equal spaces. Point E is 3 jumps from 0, so it is at 3/8.
p1b_s2	Set 3: Point F is on top of the 1. What fraction with a denominator of 4 names point F?	**4/4** (also accepted: 4 / 4, four fourths, four-fourths, 4 fourths, four quarters, 4 quarters, 4/4 mile, 4/4 of a mile, 4/4 miles, 4 / 4 mile, 4 / 4 of a mile, 4 / 4 miles)	Point F is 4 jumps of 1/4 from 0. 4 fourths make 1 whole, so F is at 4/4, the same as 1.
p1c_s2	Set 3: Find the mistake. Wen cut this number line into 3 parts and labeled the first mark 1/3. What went wrong? [The 3 parts are not the same length. Thirds must be equal parts. / Nothing is wrong. The first mark is one third. / He should label it one fourth. / He should label it 3.]	**The 3 parts are not the same length. Thirds must be equal parts.**	Thirds must be 3 equal parts. The parts on this line are different lengths, so the first mark is not 1/3.

COUNTING PAST 1: JUMPS TO THE CREEK (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
u1a_s0	Set 1: What fraction names point G? Count every jump from 0.	**5/3** (also accepted: 5 / 3, five thirds, five-thirds, 5 thirds, 5/3 mile, 5/3 of a mile, 5/3 miles, 5 / 3 mile, 5 / 3 of a mile, 5 / 3 miles)	Each whole has 3 equal spaces. Point G is 5 jumps of 1/3 from 0, so it is at 5/3.
u1b_s0	Set 1: Fill in the numerator. 2 =? over 4. How many fourths make 2 wholes?	**8** (also accepted: eight, 8/4, 8 fourths)	Each whole has 4 fourths. 2 wholes have 4 + 4 = 8 fourths, so 2 = 8/4.

ID	Question	Answer	Why (explain to student)
u1c_s0	Set 1: Which explanation is correct? Why is 3/3 at the same point as 1? [Three jumps of one third cover the whole space from 0 to 1. / Because 3 is bigger than 1. / Because three thirds is always less than 1. / Because 3 minus 3 is 0.]	**Three jumps of one third cover the whole space from 0 to 1.**	3 jumps of 1/3 cover the whole space from 0 to 1, so 3/3 lands on 1.
u1a_s1	Set 2: What fraction names point H?	**3/2** (also accepted: 3 / 2, three halves, three-halves, 3 halves, 3/2 mile, 3/2 of a mile, 3/2 miles, 3 / 2 mile, 3 / 2 of a mile, 3 / 2 miles)	Each whole has 2 equal spaces, called halves. Point H is 3 jumps of 1/2 from 0, so it is at 3/2.
u1b_s1	Set 2: Fill in the numerator. 2 =? over 6. How many sixths make 2 wholes?	**12** (also accepted: twelve, 12/6, 12 sixths)	Each whole has 6 sixths. 2 wholes have 6 + 6 = 12 sixths, so 2 = 12/6.
u1c_s1	Set 2: Find the mistake. Zuri says 5/4 is between 0 and 1 on the number line. What went wrong? [Four fourths already reach 1, so five fourths is one jump past 1, between 1 and 2. / Nothing is wrong. Five fourths is between 0 and 1. / Five fourths is past 5. / Five fourths is exactly 0.]	**Four fourths already reach 1, so five fourths is one jump past 1, between 1 and 2.**	4 fourths reach 1. 5/4 is 1 more jump, so it is between 1 and 2.
u1a_s2	Set 3: What fraction names point J?	**7/4** (also accepted: 7 / 4, seven fourths, seven-fourths, 7 fourths, seven quarters, 7 quarters, 7/4 mile, 7/4 of a mile, 7/4 miles, 7 / 4 mile, 7 / 4 of a mile, 7 / 4 miles)	Each whole has 4 equal spaces. Point J is 7 jumps of 1/4 from 0, so it is at 7/4.
u1b_s2	Set 3: Write the whole number 4 as a fraction with a denominator of 1.	**4/1** (also accepted: 4 / 1)	A denominator of 1 means each whole is 1 part. 4 wholes are 4 of those parts: 4/1.
u1c_s2	Set 3: Which explanation is correct? Where is 6/3 on a number line? [At 2, because 3 thirds make each whole, and 6 thirds make 2 wholes. / At 6, because the numerator is 6. / At 3, because the denominator is 3. / Between 0 and 1, because it is a fraction.]	**At 2, because 3 thirds make each whole, and 6 thirds make 2 wholes.**	Each whole has 3 thirds. 6 thirds is 3 + 3, which is 2 wholes, so 6/3 is at 2.

SAME SPOT, DIFFERENT NAMES (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1a_s0	Set 1: The top line is in thirds and the bottom line is in sixths. What fraction on the bottom line names the same point as 2/3?	**4/6** (also accepted: 4 / 6, four sixths, four-sixths, 4 sixths)	Follow the dashed line down. 2/3 lines up with the fourth mark on the sixths line, so 2/3 = 4/6.
e1b_s0	Set 1: Fill in the numerator. 1/2 =? over 8.	**4** (also accepted: four, 4/8, 4 eighths)	1/2 is halfway. On a line cut into 8 equal parts, halfway is 4 jumps, so 1/2 = 4/8.

ID	Question	Answer	Why (explain to student)
e1c_s0	Set 1: Find the mistake. Tomas says 2/4 and 2/3 are equivalent because they both have a numerator of 2. What went wrong? [They do not land on the same point. Two thirds is farther from 0 than two fourths. / Nothing is wrong. They are equivalent. / Two fourths is farther from 0 than two thirds. / Only fractions with the same denominator can be compared.]	**They do not land on the same point. Two thirds is farther from 0 than two fourths.**	Equivalent fractions must land on the same point. 2/4 is at the halfway point, but 2/3 is farther from 0.
e1a_s1	Set 2: The top line is in fourths and the bottom line is in eighths. What fraction on the bottom line names the same point as 3/4?	**6/8** (also accepted: 6 / 8, six eighths, six-eighths, 6 eighths)	The dashed line from 3/4 lands on the sixth mark of the eighths line, so 3/4 = 6/8.
e1b_s1	Set 2: Fill in the numerator. 1/3 =? over 6.	**2** (also accepted: two, 2/6, 2 sixths)	Cut each third into 2 equal parts to get sixths. 1/3 becomes 2 sixths, so 1/3 = 2/6.
e1c_s1	Set 2: Which explanation is correct? Why is 1/2 equal to 3/6? [Both fractions land on the same point, halfway between 0 and 1. / Because 3 and 6 are bigger numbers. / Because 1 plus 2 is 3. / They are not equal. Three sixths is bigger.]	**Both fractions land on the same point, halfway between 0 and 1.**	1/2 and 3/6 both land exactly halfway between 0 and 1, so they are equivalent.
e1a_s2	Set 3: The top line is in halves and the bottom line is in sixths. What fraction on the bottom line names the same point as 1/2?	**3/6** (also accepted: 3 / 6, three sixths, three-sixths, 3 sixths)	Halfway on the sixths line is 3 jumps, so 1/2 = 3/6.
e1b_s2	Set 3: Fill in the numerator. 6/8 =? over 4.	**3** (also accepted: three, 3/4, 3 fourths)	Every 2 eighths make 1 fourth. 6 eighths is 3 groups of 2, so 6/8 = 3/4.
e1c_s2	Set 3: Which pair of fractions is NOT equivalent? [1/3 and 3/6 / 1/2 and 4/8 / 2/3 and 4/6 / 3/4 and 6/8]	**1/3 and 3/6**	1/3 is one third of the way to 1, but 3/6 is halfway. They are at different points, so they are not equivalent.

CHEESE BLOCKS: EQUAL AMOUNTS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1a_s0	Set 1: Zuri has 3/4 of a cheese block. Wen has a same-size block cut into 8 equal pieces. How many eighths must Wen take to have the same amount as Zuri?	**6** (also accepted: six, 6/8, 6 eighths, 6 pieces)	Each fourth splits into 2 eighths. 3 fourths make 3 × 2 = 6 eighths, so Wen takes 6 pieces: 3/4 = 6/8.
d1b_s0	Set 1: The strip shows 4/6 shaded. What fraction with a denominator of 3 names the same amount?	**2/3** (also accepted: 2 / 3, two thirds, two-thirds, 2 thirds)	The strip has 6 equal parts with 4 shaded, which is 4/6. Every 2 sixths make 1 third, so 4 sixths are 2/3.
d1c_s0	Set 1: Which explanation is correct? Tomas says 4/8 of a block is more cheese than 1/2 of the same block, because 4 pieces is more than 1 piece. [He is wrong. The eighths are smaller pieces, and 4 eighths cover the same amount as 1 half. / He is right. More pieces always means more cheese. / He is wrong, because one half is more than four eighths. / He is right, because 8 is bigger than 2.]	**He is wrong. The eighths are smaller pieces, and 4 eighths cover the same amount as 1 half.**	Eighths are smaller pieces than halves. 4 small eighths cover exactly the same amount as 1 half, so 4/8 = 1/2.

ID	Question	Answer	Why (explain to student)
d1a_s1	Set 2: Two steps! A cheese block is cut into 6 equal pieces. Tomas sells 3 pieces. What fraction with a denominator of 2 tells how much of the block he sold?	**1/2** (also accepted: 1 / 2, one half, one-half, 1 half, a half)	Step 1: 3 of the 6 pieces is 3/6. Step 2: 3 sixths cover half of the block, so 3/6 = 1/2.
d1b_s1	Set 2: The strip shows 2/8 shaded. Write the same amount as fourths.	**1/4** (also accepted: 1 / 4, one fourth, one-fourth, 1 fourth, one quarter, 1 quarter, a fourth)	The strip has 8 equal parts with 2 shaded: 2/8. Every 2 eighths make 1 fourth, so 2/8 = 1/4.
d1c_s1	Set 2: Find the mistake. Zuri says 2/3 = 2/6 because the numerators match. What went wrong? [Each third splits into 2 sixths, so 2 thirds is 4 sixths, not 2 sixths. / Nothing is wrong. Two thirds equals two sixths. / Two thirds equals three sixths. / Two thirds equals two eighths.]	**Each third splits into 2 sixths, so 2 thirds is 4 sixths, not 2 sixths.**	When each third is cut into 2 sixths, the number of shaded parts doubles too: 2/3 = 4/6, not 2/6.
d1a_s2	Set 3: Two steps! A cheese block is cut into 8 equal pieces. Wen sells 2 pieces in the morning and 4 pieces in the afternoon. What fraction with a denominator of 4 tells how much of the block he sold?	**3/4** (also accepted: 3 / 4, three fourths, three-fourths, 3 fourths, three quarters, 3 quarters)	Step 1: 2 + 4 = 6 pieces sold, which is 6/8. Step 2: every 2 eighths make 1 fourth, so 6/8 = 3/4.
d1b_s2	Set 3: Mrs. Jensen cuts 3 whole blocks of cheese in half. Fill in the numerator: 3 =? over 2.	**6** (also accepted: six, 6/2, 6 halves)	Each whole block has 2 halves. 3 blocks have 2 + 2 + 2 = 6 halves, so 3 = 6/2.
d1c_s2	Set 3: Which strip shows the same amount as 2/4? [4 out of 8 equal parts shaded / 2 out of 8 equal parts shaded / 2 out of 3 equal parts shaded / 4 out of 6 equal parts shaded]	**4 out of 8 equal parts shaded**	2/4 is half of the strip. 4 out of 8 equal parts is also half, so it shows the same amount.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 12	**24**	2 points if matched on the first try.
match_s0_2	Set 1: Match 13	**26**	2 points if matched on the first try.
match_s0_3	Set 1: Match 34	**68**	2 points if matched on the first try.
match_s0_4	Set 1: Match 23	**46**	2 points if matched on the first try.
match_s0_5	Set 1: Match 44	**1 whole**	2 points if matched on the first try.
match_s1_1	Set 2: Match number line	**numbers in order at equal spaces on a line**	2 points if matched on the first try.
match_s1_2	Set 2: Match endpoint	**a point at the end of a line segment**	2 points if matched on the first try.
match_s1_3	Set 2: Match numerator	**the top number: how many parts we count**	2 points if matched on the first try.
match_s1_4	Set 2: Match denominator	**the bottom number: how many equal parts in one whole**	2 points if matched on the first try.
match_s1_5	Set 2: Match equivalent fractions	**fractions that name the same amount**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s2_1	Set 3: Match 63	**2 wholes**	2 points if matched on the first try.
match_s2_2	Set 3: Match 54	**between 1 and 2**	2 points if matched on the first try.
match_s2_3	Set 3: Match 38	**between 0 and 1**	2 points if matched on the first try.
match_s2_4	Set 3: Match 22	**1 whole**	2 points if matched on the first try.
match_s2_5	Set 3: Match 41	**4 wholes**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: the red barn is in wisconsin	**The red barn is in Wisconsin.**	Start with a capital T. Wisconsin is the name of a state, so it gets a capital W. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: wen zuri and tomas set the posts	**Wen, Zuri, and Tomas set the posts.** (also accepted: Wen, Zuri and Tomas set the posts.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s1	Set 2: Fix the sentence: where is the water trough	**Where is the water trough?**	Start with a capital W. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: mrs. jensen sells milk cheese and butter	**Mrs. Jensen sells milk, cheese, and butter.** (also accepted: Mrs. Jensen sells milk, cheese and butter.)	Mrs. and Jensen both get capital letters, commas go between the things in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: the farm is near madison	**The farm is near Madison.**	Start with a capital T. Madison is the name of a city, so it gets a capital M. End the telling sentence with a period.
fs2_s2	Set 3: Fix the sentence: how many posts did zuri set	**How many posts did Zuri set?**	Start with a capital H. Zuri is a name, so it gets a capital Z. This sentence asks something, so it ends with a question mark.

BOSS BATTLE: THE LAST POST (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Two steps! The lane is 1 mile long. Mrs. Jensen puts a fence post every 1/8 of a mile, with a post at the barn (0) and a post at the gate (1). How many posts does she need in all?	**9** (also accepted: nine, 9 posts)	Step 1: posts every 1/8 mile cut the lane into 8 equal spaces. Step 2: 8 spaces need 9 posts, because there is a post at 0 and at 1. Count them: 0, 1/8, 2/8, and so on up to 8/8.
bb2_s0	Set 1: Find the mistake. Tomas says the water trough at 3/4 mile and the salt lick at 6/8 mile are in different spots. What went wrong? [They are in the same spot. Three fourths equals six eighths. / Nothing is wrong. Six eighths is farther from the barn. / Nothing is wrong. Three fourths is farther from the barn. / Six eighths is past the gate.]	**They are in the same spot. Three fourths equals six eighths.**	Every fourth splits into 2 eighths, so 3/4 = 6/8. Both are the same point on the lane.

ID	Question	Answer	Why (explain to student)
bb1_s1	Set 2: Two steps! Posts are set every 1/6 of a mile. Zuri stops at the 4th post after the barn. What fraction with a denominator of 3 names her spot?	**2/3** (also accepted: 2 / 3, two thirds, two-thirds, 2 thirds, 2/3 mile, 2/3 of a mile, 2/3 miles, 2 / 3 mile, 2 / 3 of a mile, 2 / 3 miles)	Step 1: the 4th post after the barn is at 4/6 mile. Step 2: every 2 sixths make 1 third, so 4/6 = 2/3.
bb2_s1	Set 2: Three of these fractions name the same point. Which one does NOT? [3/8 / 1/2 / 2/4 / 4/8]	**3/8**	1/2, 2/4, and 4/8 all land halfway. 3/8 lands before the halfway point.
bb1_s2	Set 3: Two steps! Posts are set every 1/4 of a mile. Tomas walks from the barn to the 3rd post, then walks back 1 post. How far from the barn is he? Write your answer in eighths.	**4/8** (also accepted: 4 / 8, four eighths, four-eighths, 4 eighths, 4/8 mile, 4/8 of a mile, 4/8 miles, 4 / 8 mile, 4 / 8 of a mile, 4 / 8 miles)	Step 1: he walks to the 3rd post, 3/4, then back 1 post to 2/4. Step 2: each fourth is 2 eighths, so 2/4 = 4/8.
bb2_s2	Set 3: Which explanation is correct? Why does 1/4 equal 2/8? [Each fourth splits into 2 equal eighths, so one fourth covers the same length as two eighths. / Because 2 plus 8 is 10. / Because 1 fourth is smaller than 2 eighths. / They are not equal.]	**Each fourth splits into 2 equal eighths, so one fourth covers the same length as two eighths.**	Each fourth splits into 2 equal eighths, so 1/4 covers the same length as 2/8.

BOSS BATTLE: CLOSING THE GATE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: What fraction names point M?	**5/3** (also accepted: 5 / 3, five thirds, five-thirds, 5 thirds, 5/3 mile, 5/3 of a mile, 5/3 miles, 5 / 3 mile, 5 / 3 of a mile, 5 / 3 miles)	Each whole has 3 equal spaces. Point M is 5 jumps of 1/3 from 0: 5/3.
bb4_s0	Set 1: Which explanation is correct? Are 2/2 and 4/4 at the same point on a number line? [Yes. Both equal 1 whole, so they land on 1. / No. Four fourths is farther because 4 is bigger. / No. Two halves is farther because halves are bigger pieces. / Yes, but only because both numerators are even.]	**Yes. Both equal 1 whole, so they land on 1.**	2/2 and 4/4 both have matching numerators and denominators, so each one equals 1 whole and they land on the same point.
bb3_s1	Set 2: Fill in the numerator. 3 =? over 4.	**12** (also accepted: twelve, 12/4, 12 fourths)	Each whole has 4 fourths. 3 wholes have 4 + 4 + 4 = 12 fourths, so 3 = 12/4.
bb4_s1	Set 2: Find the mistake. On a number line cut into thirds, Zuri counted 2 tick marks starting with the tick at 0, and marked that tick as 2/3. What went wrong? [She counted the tick at 0. Two thirds is 2 jumps from 0, at the second mark after 0. / Nothing is wrong. That tick is two thirds. / Two thirds is past 1. / Two thirds is at 0.]	**She counted the tick at 0. Two thirds is 2 jumps from 0, at the second mark after 0.**	Start counting at 0 and count jumps. 2/3 is 2 jumps from 0, the second mark after 0.

ID	Question	Answer	Why (explain to student)
bb3_s2	Set 3: What fraction with a denominator of 4 names point N?	**3/4** (also accepted: 3 / 4, three fourths, three-fourths, 3 fourths, three quarters, 3 quarters, 3/4 mile, 3/4 of a mile, 3/4 miles, 3 / 4 mile, 3 / 4 of a mile, 3 / 4 miles)	Point N is 6 jumps of 1/8, which is 6/8. Every 2 eighths make 1 fourth, so 6/8 = 3/4.
bb4_s2	Set 3: Find the mistake. Wen wrote 1 = 1/4 because there is 1 whole. What went wrong? [One whole cut into fourths has 4 fourths, so 1 equals four fourths. / Nothing is wrong. One equals one fourth. / One equals four over one. / One equals zero fourths.]	**One whole cut into fourths has 4 fourths, so 1 equals four fourths.**	One whole cut into fourths has 4 fourths, so 1 = 4/4. 1/4 is only one part.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W17.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 5 of 8 slices have pepperoni	5/8 of the pizza has pepperoni.
2	Line from 0 to 1 in 2 equal parts; name the middle point	The middle point is 1/2.
3	Tick marks on a 0-to-1 line with 4 equal spaces	5 tick marks (0, 1/4, 2/4, 3/4 and 1).
4	Wen takes 1 of 2 pieces; Zuri takes 2 of 4 pieces	They have the same amount: 1/2 = 2/4 of same-size blocks. Drawings should show equal halves.
5	Fraction equal to 1 whole	4/4

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Number line 0 to 1 in thirds; post at 2/3	Ticks labeled 0, 1/3, 2/3 and 3/3 (or 1), with P at 2/3, two equal jumps from 0.
2	Posts every 1/8 mile; 6th post in eighths, then fourths	6/8 of a mile, which is the same point as 3/4 of a mile.
3	Wen says 2/6 = 2/3 because the numerators match	Sixths are smaller pieces than thirds, so 2/6 is less than 2/3. Equivalent fractions must land on the same point. 2/3 = 4/6.
4	Why 7/6 is not between 0 and 1	It takes 6 jumps of 1/6 to reach 1, so 7 jumps go past 1. 7/6 is between 1 and 2.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Fourths line over eighths line; circle matching pairs	0 = 0/8, 1/4 = 2/8, 2/4 = 4/8, 3/4 = 6/8, 4/4 = 8/8.